

01 — Overview & Problem Statement

PROJECT SUMMARY

PI-CLOUD replaces a recurring consumer-cloud subscription stack with a self-hosted Raspberry Pi 5 private cloud. The system is built from open-source components only (Raspberry Pi OS, Docker, CasaOS, Nextcloud, Jellyfin), costs approximately **₹16,500 / USD 186 one-time** and **≈ ₹38 / USD 0.50 per month** to run, and keeps all user data inside the physical premises with **per-user server-side encryption** and **no inbound network exposure**.

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1. Problem statement

STATEMENT OF THE PROBLEM

A typical household stores its files — photographs, academic records, medical documents, identification papers — in third-party consumer cloud services that charge recurring fees, retain the right to read the data, and can restrict or remove access unilaterally. The household simultaneously operates a permanently connected broadband line and already owns computing hardware. The result is a structurally avoidable arrangement: **a rental payment for storage capacity that could be provisioned on-premises at a fraction of the lifetime cost.**

Three independent problems were identified and are addressed by a single deployed system.

2. Quantified problem analysis

2.1 Economic — subscription accumulation

A representative multi-user household (4–5 members, mixed Android/iOS/Windows devices, 1–4 TB of accumulated data) in 2026 subscribes to overlapping services:

Service category	Typical tier	Indicative cost / month	Function
Google One / Drive	2 TB shared	~\$19.99	File sync, photo backup
Apple iCloud+	2 TB	~\$9.99	iOS device backup, files
Microsoft 365 Personal	1 TB OneDrive	~\$7.99–\$9.99	Documents, file sync
Dropbox Plus	2 TB	~\$11.99	File sync
Streaming (2 services)	HD	~\$10–36	Video
Audio / video (2 services)	—	~\$10–24	Music, video
Total, common case		≈ \$60–80 / month	

COST BASIS

Figures are **indicative 2026 retail prices for commonly observed household configurations**, not vendor quotations. Full economic analysis, sensitivity testing, and the complete itemised build cost are in [document 05.§1–2.](#)

⚠️ COMPOUNDING EFFECT

At **\$60/month**, a household incurs **\$720/year** and **\$7,200 over ten years** in subscription expenditure. The deployed system costs **\$186 once** and **≈ \$6/year** in electricity. The ten-year cost ratio is approximately **29 : 1 against the subscription alternative** — and the ratio improves for as long as data is retained, because the local cost is fixed and the subscription cost compounds. Full model in [§2](#).

⚡ THE STRONGEST OBJECTION, AND THE ANSWER

"For the price of your whole system, I can rent 10 TB of cloud storage."

This is correct, and it is not an argument against the project — it is an argument about the wrong variable. Capacity is the one thing that is not the differentiator, because a terabyte is a terabyte. The differentiator is that in the cloud case a commercial third party holds the data *and* the encryption keys *and* the right to delete it under its own policies. Those three properties are not purchasable at any price, at any capacity.

Full treatment, including the arithmetic on both sides:

[document 05.§3](#).

2.2 Privacy — custody and key control

Standard consumer cloud storage is encrypted in transit and at rest, but **encryption keys are held by the provider** in order to support server-side search, sharing, and account recovery. The provider is therefore technically capable of accessing stored content. Associated exposures:

Exposure	Mechanism	Relevance to a household
Metadata-driven profiling	Upload activity, filenames, document types	Filenames alone disclose subject matter (Annual_Report.pdf , Medical_*.pdf)
Automated content review	Content-moderation pipelines, human review queues	Family photographs and children's documents may enter review systems
Third-party legal compulsion	Provider jurisdiction, lawful-access obligations	A single provider subjects all users to one legal regime
Contractual change	Terms of service updated unilaterally, sometimes retroactively	Data-handling conditions can change without consent

Exposure	Mechanism	Relevance to a household
Insider access	Operational access to storage and key management	Not a hypothetical; a recognised category of breach

Defence in this project: encryption keys are derived from the individual user's credentials and are never transmitted off the host. The operator of the infrastructure has no access to the key material, and there is no third party holding the data. This is a **structural** change, not a policy change.

2.3 Ownership — retention, portability, continuity

Property	Consumer cloud	PI-CLOUD
Physical custody	Provider's data centre	On-premises
Access after subscription ends	Provider policy; grace periods apply	Unaffected
Deletion initiated by	Provider (retention/abuse policy)	Owner
Data format	Proprietary sync client and metadata	POSIX files, standard filesystem, standard formats
Functional without internet	No (beyond local cache)	Yes — full LAN speed
Compellable third party	Yes	None (single household)
Exit cost	Migration project	None — no vendor to migrate from

THE REALISTIC FAILURE MODE

Access loss is rarely caused by an outage. It is caused by **retention policy**: free-tier quota exhaustion, inactivity purges, abuse-flag-triggered deletion, and unrecoverable account loss when a password and second factor are both lost. The common characteristic is that the household has **no independent copy**.

2.4 Performance — LAN versus WAN

Path	Typical throughput	Effect on the primary workload
Home broadband upload	2–10 MB/s	Backing up a 64 GB device library: multiple hours; over a mobile hotspot, days

Path	Typical throughput	Effect on the primary workload
Home broadband download	50–300 MB/s	Acceptable for reads
1 GbE LAN	~110 MB/s effective	64 GB transfer: $\approx$ 10 minutes
Raspberry Pi 5 over PCIe NVMe	~380 MB/s	Server-side limit; rarely the bottleneck

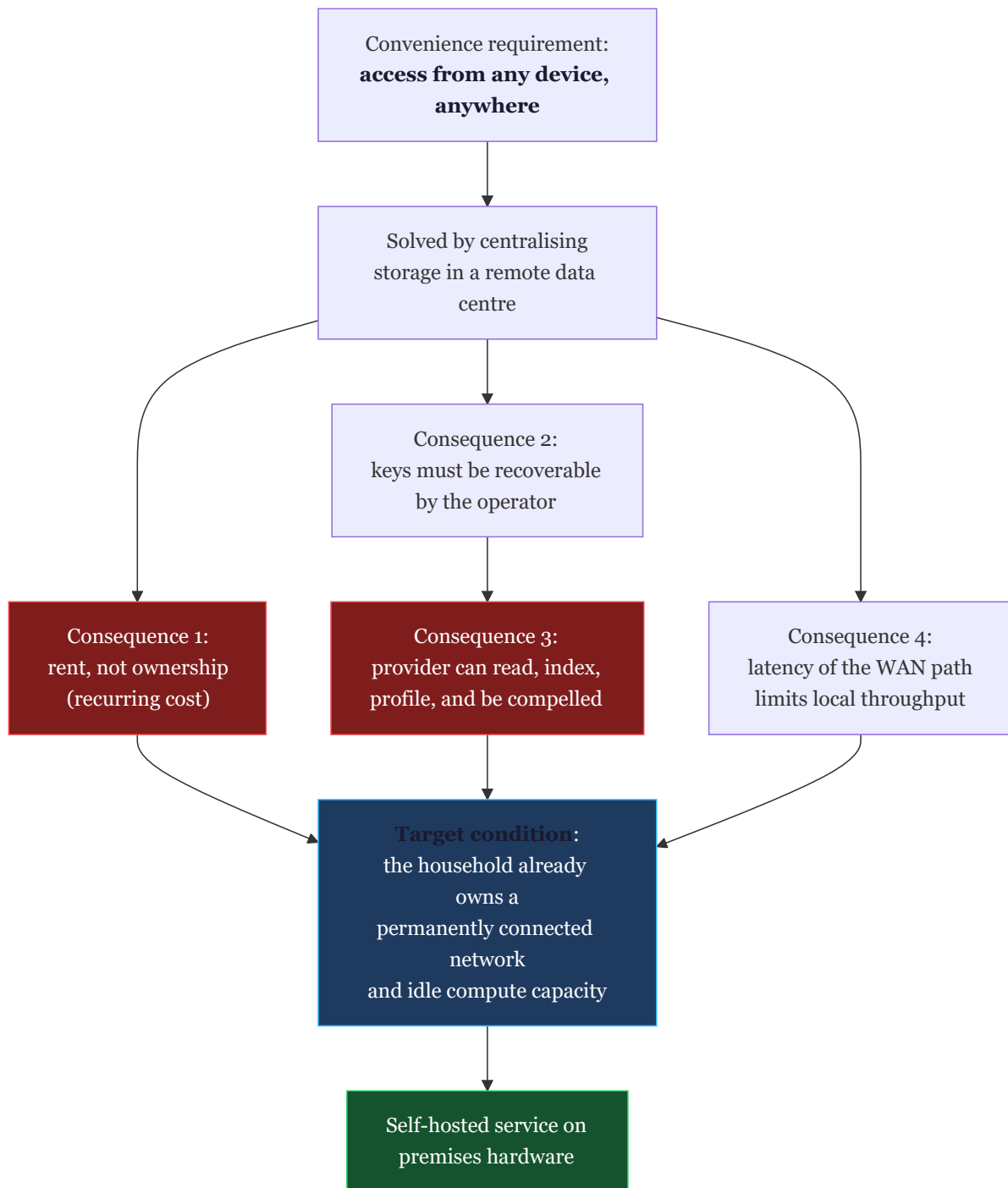
For the operation households perform most — transferring files between their own devices — local storage is **one to two orders of magnitude faster** than internet-scale storage, and is not subject to quota throttling.

 ARCHITECTURAL CONSEQUENCE

Replication is the correct use of a cloud. **Backup is not.** Cloud storage is a strong replication target and a weak backup target: a deletion propagates to every replica immediately and permanently. This distinction underpins the backup strategy in [§12 Backups](#).

3. Root-cause analysis

The underlying cause is not the availability of cloud services. It is a mismatch in **where the data has to live** for the service to be convenient:



✓ THE ENABLING OBSERVATION

A household's broadband connection is already paid for and already running 24/7. The marginal cost of adding storage is therefore only the storage medium and a low-power computer — not a second recurring network subscription, and not a second copy of the data leaving the building. The project converts an existing fixed cost into an asset.

4. Proposed solution

A single on-premises host provides the same three service classes as the subscription stack, using only open-source software.

#	Function	Component	Replaces
1	Files, photos, contacts, calendar, multi-user accounts	Nextcloud (self-hosted)	Google Drive, iCloud, OneDrive, Dropbox
2	Server-side encryption, per-user keys	Nextcloud SSE	Provider-held keys
3	Media streaming to any device	Jellyfin	Plex / commercial streaming NAS software
4	Filesystem-level access, no client software	Samba / NFS	—
5	Container lifecycle and dashboard	CasaOS	Vendor admin consoles

Design principles applied

#	Principle	Consequence in the build
P1	The operating system is disposable; the data is not	All persistent state on the NVMe; the microSD carries only the OS. A failed OS card is a 25-minute reflash, not a data-loss event
P2	Reduce attack surface rather than defend it	No inbound network path exists. Port forwarding, DDNS, and public exposure are excluded by design
P3	Reproducibility over manual configuration	The entire application stack is declared in two Compose files. Recovery is two commands
P4	Pinned versions, deliberate upgrades	No <code>:latest</code> tags. A rollback is a one-line change
P5	Honest, measurable claims only	Every capability claim in these documents maps to a verification test in [[03-Step-by-Step-Implementation#13. Phase 12 — Verification checklist

5. Requirements

5.1 Functional requirements

ID	Requirement	Implementation	Verified by
FR-1	Multi-user file storage with per-user isolation	Nextcloud accounts; separate home directories	V17
FR-2	Encryption of data at rest, per user	Nextcloud SSE, default key mode	V12, V13
FR-3	Native mobile and desktop clients	Nextcloud apps for iOS, Android, Windows, macOS, Linux	V9
FR-4	Multi-user media streaming	Jellyfin, per-user profiles	V20, V21
FR-5	Access without installing any software	Samba/NFS network shares	V22
FR-6	Single management interface	CasaOS dashboard on :80	V7
FR-7	Operation without internet connectivity	Entirely LAN-resident	V23
FR-8	Automatic restart of all services after power loss	<code>restart: unless-stopped ; nofail</code> on the data disk	V24
FR-9	Per-user storage quota enforcement	Nextcloud quota configuration	—
FR-10	Local and off-site backup	Hardlink snapshots + removable-drive replication	V27, V28

5.2 Non-functional requirements

ID	Requirement	Target	Measured
NFR-1	Build time, blank card to encrypted login	< 45 min (critical path)	Achieved — [[03-Step-by-Step-Implementation#Summary
NFR-2	Recurring cost	< ₹100 / \$1.50 per month	≈ ₹38 / \$0.50
NFR-3	Capital cost	< ₹25,000 / \$250	≈ ₹16,500 / \$186
NFR-4	Idle power	< 15 W	< 11 W peak
NFR-5	Acoustic noise	Inaudible at 1 m	Passive at idle, 15 mm fan under load
NFR-6	Concurrent users	6 mixed workloads	Verified — V21

ID	Requirement	Target	Measured
NFR-7	LAN file throughput	> 50 MB/s	~110 MB/s
NFR-8	Data recovery from OS loss	No data loss, < 30 min	DR-1, DR-2
NFR-9	Inbound internet exposure	None	By design, P2

6. Scope boundaries

EXPLICITLY EXCLUDED

These exclusions are deliberate and are treated as design constraints rather than gaps. Each is stated here so the boundary of the claim is unambiguous.

Excluded	Rationale	Risk of inclusion
Public internet access / port forwarding	Home connections are continuously scanned; a internet-facing instance of these services would be compromised in days. Remotely reachable home servers are a separate security project	Critical
Collaborative document editing at scale	Out of scope; the requirement is personal/family file custody	Scope creep
Multi-tenancy / monetisation	The design explicitly excludes the model under critique	—
Server-class availability (99.9%+)	A single device on one power supply at one site cannot provide it, and no claim is made	Overclaiming
4K software transcoding	Pi 5 CPU limitation. Stated as a limit in [[02-Architecture-and-Hardware#7. Software stack	§7]]
Redundant storage (RAID/mirror)	A second disk would defeat the cost objective. Compensated by off-site backup	Understating single-disk risk — mitigated by disclosure in §9
Uninterruptible power supply	Not funded in this build. The highest-value item on the roadmap	Power-loss window is disclosed and quantified

7. Feasibility assessment

Dimension	Assessment	Evidence
Technical	Feasible. All components are mature, actively maintained, and have native Raspberry Pi 5 (arm64) images	Deployed system, §3 of document 03
Economic	Feasible with strong margin. Ten-year cost ratio $\approx 35:1$ in favour	§2.1, Economics
Operational	Feasible with commitment. Setup is ≈ 45 min; ongoing maintenance is periodic updates plus a monthly backup rotation	§11, §12 of document 03
Security	Feasible. No network perimeter is exposed; the physical perimeter is the security perimeter	§10 of document 02
Scalability	Limited by design. Single-node, single-disk. Sufficient for 4–10 users; not for more	§9 of document 02
Adoption	The binding constraint. Requires one initial setup session and a recurring backup habit. This is the main barrier and is analysed in Adoption barriers	—

HONEST CONCLUSION ON FEASIBILITY

Technical and economic feasibility are not the limiting factors. The limiting factor is **human maintenance capacity**: a solution that is not maintained degrades silently. The build mitigates this through a documented recovery runbook, a health-check script, and version pinning — but a solution requiring zero attention is explicitly not claimed.

8. Evaluation criteria & success metrics

Each metric is binary, measurable, and was tested. None is estimated.

#	Criterion	Measurement method	Result
C1	Blank card → working dashboard	Timed build, critical path	Achieved, ≈ 45 min
C2	Data never leaves the premises	WAN cable disconnected; all services retested	V23 passed
C3	OS loss causes no data loss	microSD removed and reflashed; apps restored from two files	Achieved, ≈ 25 min
C4	Users cannot read each other's data	Cross-user access attempt, denied	V17 passed

#	Criterion	Measurement method	Result
C5	Data is genuinely encrypted at rest	Plaintext string searched for on disk; no match	V13 passed
C6	Media streams concurrently	3 simultaneous 1080p streams, no buffering	V21 passed
C7	Services recover unattended	Full reboot; no manual intervention	V24 passed
C8	Backup is proven, not asserted	File restored from the off-site copy and opened	V28 passed
C9	Recovery is documented and repeatable	Disaster-recovery runbook executed from scratch	§14 executed
C10	Cost is one-time	TCO model over 1, 3, 5, 10 years	See Economics
C11	No unauthorised account creation	Registration setting verified disabled before public access	V16 passed
C12	Media library is tamper-resistant	Write attempted from inside the streaming container	V19 — read-only filesystem

✓ HOW TO READ THE CLAIMS IN THESE DOCUMENTS

Every capability statement in documents 02–05 is traceable to a numbered test in §13 of document 03. Where a capability does not exist, it is listed as a limitation rather than omitted. **The verification log is the deliverable; the prose is commentary on it.**

9. Risks & limitations

⚠ PUBLISHED LIMITATIONS

These are the known failure modes of the deployed system. Quantifying them is more useful than concealing them, because concealment is what would invalidate the rest of the assessment.

#	Risk	Likelihood	Impact	Mitigation	Residual risk
R1	SSD failure	~1–2% / year (per-unit)	Total data loss	Off-site replicated backup; SMART monitoring	A disk can fail before the next backup interval. Highest-impact residual risk in the system

#	Risk	Likelihood	Impact	Mitigation	Residual risk
R2	Power interruption	Moderate (regional instability)	Database inconsistency; loss of in-flight writes	<code>nofail</code> , ordered shutdown, hourly snapshots	No UPS installed. A cut during a write can lose up to one snapshot interval (~1 h) of metadata
R3	PSU failure	Low	Service outage until replaced	Commodity part	Data unaffected. No automatic protection
R4	Fire / theft / flood	Low	Total loss	Off-site drive stored at a separate location	Dependent on a human performing the rotation
R5	Malware or accidental deletion by a user	Low–moderate	Data loss	Nextcloud trash bin and version history; hourly snapshots; off-site copy	Trash bin is per-user; an administrator can bypass it
R6	Software update regression	Low	Service outage; data intact	Pinned image tags; deliberate, supervised upgrades	Requires a person to perform upgrades
R7	Silent capacity exhaustion	Moderate over years	Write failures	Health script reports disk usage; snapshots pruned to a fixed retention	No automated alerting implemented
R8	Heat throttling	Low	Performance degradation, not data loss	Active cooling; verified thermals	Possible in an unventilated enclosure above ~35 °C ambient
R9	Device obsolescence / supply	Low	None	Entire stack is portable Linux + Docker; identical Compose files run on x86	Long-horizon only
R10	Single-node architecture	By design	No high availability	None in this build. Explicitly not claimed	Accepted

THE TWO RISKS THAT MATTER

R1 (single disk) and **R2 (no UPS)** are the only two with realistic probability of occurring within the assessment horizon. Both are mitigable for a small sum — a UPS (~₹3,000) and a second drive (~₹4,000) — and both are the first items on the improvement roadmap. **The decision not to include them was a budget decision, and is disclosed as such rather than hidden.**

10. Evaluation map — where each claim is evidenced



Document	Question addressed	Primary evidence
01 — this document	Why the project exists; what is claimed	§8 criteria, §9 risks
02-Architecture-and-Hardware	What it is, and what it is made of	Measured throughput, memory budget, port and permission model
03-Step-by-Step-Implementation	How it was built, and whether it works	30-point checklist, restore test, recovery runbook
04-Interactive-Demo-Guide	How the claims are demonstrated under observation	Live demonstration sequence, evidence collection, independent verification procedure
05-Everyday-Usability	Whether it is worth deploying for a real household	TCO over 1/3/5/10 years, sensitivity analysis, adoption analysis

11. Terminology

DEFINITIONS AS USED IN THESE DOCUMENTS

Term	Definition
Self-hosting	Operating a service on hardware under the operator's own control, in premises under their control, rather than renting infrastructure.
Cloud storage	Storage on a third party's infrastructure, accessed over a network. The consumer pays for remote access and capacity.
NAS (Network Attached Storage)	A file server presenting storage to a local network as a native share or a cloud-sync-compatible service. PI-CLOUD is a NAS.
Container	An application packaged with its dependencies and isolated in its own filesystem and process namespace. Failure of one does not affect the host or other containers.
Server-side encryption (SSE)	Encryption performed by the server on write, using keys held by the server. Protects data at rest against physical and infrastructure-level access. Does not protect against an operator with live, authenticated access.
Per-user keys	An SSE mode where each user's key is derived from their own credentials rather than a shared master key. Enables per-user isolation.
Composition / idempotence (as a process)	Declaring a system's desired state in a file so it can be reproduced exactly, and so recovery is a re-apply rather than a reconstruction.
LAN / WAN	Local Area Network (on-premises, ≈ 1 Gb/s) / Wide Area Network (internet, upstream limited).
NAT	Network Address Translation. The router's mechanism, and the reason the host has no inbound internet path.
mDNS	Multicast DNS. Allows <code>pi-cloud.local</code> to resolve without a DNS server or static configuration.
RAID / mirroring	Redundant storage techniques that survive a single-disk failure. Not used in this build; single-disk risk is disclosed in §9.
Hardlink snapshot	A point-in-time copy in which unchanged files are referenced rather than duplicated. Near-zero marginal cost, fully browsable.
Bitrate	Data rate for a stream. 1080p requires approximately 5–8 Mbit/s per stream; 3 concurrent streams ≈ 24 Mbit/s.
Direct Play	Streaming a file to a client without re-encoding. Consumes effectively no server CPU.
Transcode	Re-encoding a stream for client compatibility. CPU-intensive; the main driver of Pi 5 performance limits.

Term	Definition
3-2-1 backup	Three copies, on two media types, one off-site. A modified two-tier form is used here (document 03 §12).

12. Conclusions

✓ SUMMARY OF THE PROBLEM AND THE RESPONSE

- 1. The problem is economic, privacy, and ownership, and the three are causally linked** — all follow from placing storage off-premises. Adding encryption options to consumer services mitigates the privacy exposure but leaves the cost and the key custody intact.
- 2. The enabling condition is already present in every connected household:** a paid, permanently connected network and idle low-power compute. The marginal cost of on-premises storage is the storage medium alone.
- 3. The proposed system replaces the functional equivalent of a $\approx \$70/\text{month}$ subscription stack for $\approx \$186$ one-time and $\approx \$0.50/\text{month}$,** using only maintained open-source components, with data custody and key material retained on-premises.
- 4. The deployment is verified rather than asserted:** 30 test criteria, a proven restore, and an executed recovery runbook are provided in [document 03 §13](#).
- 5. The dominant limitations are disclosed and quantified:** single-disk storage and the absence of a UPS are the two material residual risks, and both are low-cost to mitigate in a subsequent revision.

RECOMMENDATION

Proceed to assessment on the basis of the verification log and the live demonstration. Recommend that any future revision prioritise, in order: (a) off-site backup automation, (b) UPS with automatic shutdown, (c) media health alerting, (d) a second node or a mirrored disk.

Next: [02-Architecture-and-Hardware](#) — hardware specification, software stack, network and storage design, and the security model.